

Measuring cluster size

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

Measuring cluster size

1. Revive frozen stock, re-acclimate, and put through settling selection for 5 days ([[Daily growth and settling selection]] and [[Reviving strains from frozen stock]]).
2. After the fifth day's 24 h growth, pipette a random 1 mL subsample into a 1.5 mL Eppendorf tube.
3. Add 0.5 mL sterile water to each well of a 12-well plate.
4. Vortex the yeast sample gently and dilute into the water:
 - a. Microscopic populations: 1,000-fold dilution.
 - b. Macroscopic populations: 100-fold dilution.
5. Shake the well plate gently to disperse clusters evenly across the well bottom.
6. Image each well with a $\times 4$ objective, capturing cluster cross-sectional area without disrupting 3D structure. Include a physical 100 μm scale bar per image for pixel-to-micrometer conversion.

Image analysis pipeline (cluster segmentation): zipped folder containing pipeline is attached — segments whole clusters (brightfield-based, supports single field-of-view and stitched whole-well images).